MILITARY SPECIFICATION SHEET

HOSE ASSEMBLIES, BREATHING OXYGEN, LOW PRESSURE CONNECTOR TO REGULATOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the hose assemblies described herein shall consist of this document and the issue in effect of Specification MIL-H-81581.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for three types and four styles of low pressure oxygen breathing hose assemblies from connector to regulator, with or without integrated services for communication continuity.

1.2 Classification. Hose and hose assemblies covered by this specification shall be of the following types and styles, as designated by the applicable MS number, as specified (see 6.2):

Type I - Hose assembly without integrated communication cable and without connector

Type II - Hose assembly with integrated communication cable and without connector

Type III - Hose with integrated microphone switch, communication cable and without connector

Style A Assembly - Type I hose with MS22058-1 connector installed

Style B Assembly - Type I hose with MS22058-2 connector installed

Style C Assembly - Type II hose with MS22058-1 connector installed

Style D Assembly - Type III hose with MS22058-1 connector installed

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1660

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2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	-	Preservation, Methods of.
MIL-H-81581	-	Hose Assemblies, Breathing Oxygen and Air: General Specification for.

STANDARDS

MILITARY

MS22055	-	Hose Assemblies, Oxygen Breathing, Connector to Regulator.
MS22058	-	Connector, Oxygen Hose to Regulator.
MS22064	-	Clamp, Oxygen Hose.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with the general specification MIL-H-81581 and as specified herein.

3.2 Qualification. The hose furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.1.1). Unless otherwise specified, each type and style shall be qualified separately.

3.3 First article. When specified, a sample shall be subjected to first article inspection (see 4.1.2).

3.4 Design and construction. The hose assembly shall be in accordance with the general specification and with the applicable hose assembly of MS22055.

3.4.1 Oxygen mask to regulator hose connector. A fitting, when required, as shown on MS22058, shall be attached to one end of the hose by means of MS22064 clamp. The clamp shall be tightened so that a pull of at least 40 pound is required to separate the connector from the hose. One additional clamp shall be furnished for attaching the hose to the regulator.

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3.5 Performance. Unless otherwise specified herein, the performance of the hose assembly shall be as specified in the general specification, consistent with the applicable performance requirements for the applicable tests and inspections specified herein.

3.5.1 Polarity (color code) and electrical continuity. When tested as specified in the general specification, the resistance measured between all points designed to be electrically connected shall be as specified in the general specification with an allowable resistance of 0.3 ohms per foot in length of the electrically wired portion of the hose.

3.5.2 Elongation. The Type I hose, when tested as specified in the general specification, shall show an increase in length of not less than 40 percent, except for the 12 inch lengths which shall show an increase in length of not less than 30 percent. The Types II and III hoses, when tested as specified in the general specification, shall show an increase in length of not less than 5 percent. The permanent set for all hoses shall not exceed 5 percent.

3.5.3 Strength.

3.5.3.1 Tensile load. The hose, when tested as specified in the general specification, shall not separate from the end connector or the molded end.

3.5.3.2 Static load. The hose, when tested as specified in the general specification, shall not show a deflection, on the outside diameter of the hose, of more than 25 percent of the original value. The outside diameter of the hose, when checked as specified with the load removed, shall be within 5 percent of the original value.

3.5.4 Restraint cord elongation and tensile strength. When the applicable control drawing specifies a restraint cord, the restraint cord shall have a 30 percent maximum elongation and a minimum tensile strength of 70 pounds, when tested as specified in the general specification.

3.5.5 Weight. When tested as specified in the general specification, the weight of the Type I hose shall not exceed 5.0 ounces per foot of nominal specified length of hose. The weight of the Types II and III hoses shall not exceed 7.0 ounces plus 6.0 ounces per foot of nominal specified length of hose.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspections. Sampling and inspections shall be in accordance with the general specification and as specified herein.

4.1.1 Qualification inspection. The qualification inspection of the hose assembly shall consist of the following examinations and tests from the general specification:

Visual examination
Polarity (color code) and electrical continuity, Procedure I
Insulation resistance, Procedure I
Elongation, Procedure II
Delamination, Procedure I
Odor, Procedure I

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Cleanliness, Procedure I and II
Leakage, Procedure II
Flexibility,
  Type I, Procedure II
  Types II and III, Procedure IV (under environmental conditions, Procedure III)
Minimum burst pressure, Procedure II
Tensile load, Procedure IV
Static load, Procedure I
Restraint cord elongation and tensile strength
Low temperature, Procedure I
High temperature, Procedure I
Flexibility endurance, Procedure II
Ozone resistance, Procedure II
Weight, Procedure I

4.1.1.1 Qualification samples. The qualification samples shall be as specified in the general specification MIL-H-81581 with the following exception:

a. Four 36-inch long hose assemblies of each type and style for which qualification is desired.

4.1.2 First article inspection. The first article inspection of the hose assembly shall consist of the applicable examinations and tests specified in the general specification with the applicable procedure as specified herein.

4.1.2.1 First article samples. The first article sample shall consist of four hose assemblies of each style contracted for.

4.1.3 Quality conformance inspection. The quality conformance inspection shall consist of the following tests from those required by the general specification. The sample size and acceptance criteria shall be as specified in the general specification.

Visual examination
Polarity (color code) and electrical continuity, Procedure I
Insulation resistance, Procedure I
Elongation, Procedure II
Delamination, Procedure I
Odor, Procedure I
Cleanliness, Procedure I and II
Leakage, Procedure II
Flexibility,
 Type I, Procedure II
 Types II and III, Procedure IV (under environmental conditions, Procedure III)
Packaging

5. PACKAGING

5.1 Packaging. Packaging shall be in accordance with the general specification.

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6. NOTES

6.1 Intended use. The hose assembly covered by this detail specification is intended for use in supplying breathing oxygen to aircrewmen in the operation of aircraft. The hose assembly, which may or may not contain integrated communication services for aircrewmen in the operation of the aircraft, extends from the connector to the oxygen breathing regulating device.

6.2 Ordering data. Ordering data shall be in accordance with the general specification.

Custodians:
Army - AV
Navy - AS
Air Force - 11

Review activity:
Air Force - 99

Preparing activity:
Navy - AS

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